

LUNG CANCER — Screening, Dx & First Steps

#1 cancer death worldwide · Stage I 5-yr ~73–90% vs Stage IV ~0–10% · USPSTF 2021 / ACS 2023 · IM resident reference

1 · WHO TO SCREEN (asymptomatic)

Guideline	Annual low-dose CT if...
USPSTF 2021 (coverage)	Age 50–80 · ≥20 pack-yr · current smoker OR quit ≤15 yr ("A-50-80-20-15")
ACS 2023 (expanded)	Age 50–80 · ≥20 pack-yr · current OR former — no quit-time limit

Not eligible → reassess risk annually + **smoking cessation** (highest-yield). Other RFs: radon, occupational/chemical, 2nd-hand smoke, FHx (1st-deg), COPD/pulm fibrosis, prior CA.

2 · SYMPTOMATIC PATHWAY

Sx: cough, hemoptysis, dyspnea, chest pain; advanced → wt loss, malaise, bone pain (mets), HA/seizure (brain).

- **CXR first.** Not concerning → risk-assess (hemoptysis/persistent sx still → CT). Concerning → CT chest ± A/P (contrast for nodes).

CT red flags: >15 mm · irregular/spiculated · thick-walled cavity · solid component in GGO · extrapulmonary lesions · interval growth.

Tissue is the issue. Core/excisional > FNA (preserves tissue for molecular).

3 · WHERE TO BIOPSY

Biopsy the MOST ADVANCED safe lesion — one sample = diagnosis + staging (distant met = stage IV).

Site	Service
Peripheral lung	IR / Surgery (CT-guided)
Endobronchial	Pulmonology (bronch)
Liver	IR (CT/US-guided)
Node / soft tissue	Gen surg (excis.) or IR (core)
Bone	Ortho / IR

4 · STAGING THE CONFIRMED CANCER

Domain	Test
Distant	CT C/A/P w/ contrast OR PET-CT
CNS	MRI brain ±contrast (PET poor — brain always lights up)
Nodes	Bronch ± mediastinoscopy
Surgical	PFTs (if resection candidate)
Labs/molec	CBC, BMP+Ca, LFTs; PD-L1; EGFR/ALK/ROS1

~50% of never-smokers have a targetable driver — molecular testing is mandatory; changes early-stage care too.

5 · SCLC vs NSCLC

	SCLC (10–15%)	NSCLC (80–85%)
Character	Smoking, aggressive, brain mets, paraneoplastic	Adeno/squam/other; molecular-driven
Stage	Limited vs Extensive	I–IV
Tx	Chemo+RT; limited may resect; PCI	Early: resect/RT; adv: systemic
5-yr	Ltd ~10–13% / Ext ~1–2%	I ~73–90% → IV ~0–10%

6 · SYSTEMIC THERAPY

Intent: neoadjuvant (pre-op shrink) / adjuvant (post-op clear residual) / palliative (control, not curative ≠ pall care svc).

Class	Examples	Tox
Chemo	Carbo/cis, pemetrexed, etoposide, taxanes, lurbinectedin	N/V, cytopenias, neuropathy, AKI
Immuno	Pembro, nivo±ipi, atezo, durva	The "-itises" (any organ, may be late)
Targeted (-tinib)	Osimertinib(EGFR), alectinib(ALK), entrectinib(ROS1)	Rash, diarrhea; rare pneumonitis/QTc

2023–25: Resectable NSCLC → periop immunotherapy (CheckMate 816, KEYNOTE-671); EGFR-mut resected → adjuvant **osimertinib** (ADAURA, OS benefit).